



The Cyanogenmod allowed users to customize the appearance and functionality of their smartphone

The latest release is the 8.0 'Oreo' update.

Android does not include an X server like Xorg. The way an x server works is that it is able to communicate with the local computer to display a graphical interface and receive input from the user.

While that's how it works on

desktops, Android uses something Google developed called SurfaceFinger to draw the GUI on to what it calls surfaces. Of course, because it's Linux based, there are many open source Android projects that allow X-servers to be installed on an Android device. Though why you'd want a desktop on a mobile is beyond us!

Thanks to an active open source community of hobbyists and developers, there are many versions of Android systems that tweak various features. For example, the ability to overclock or under/overvolt a device's processor could be part of a custom firmware, or just a totally new UI, etc. Cyanogenmod was a good example of an open source project that allowed tweaking of the Android system.

Of course, many manufacturers will immediately void your warranty if you root your device (get root access to the kernel) in order to install such custom firmware. So be aware of that. There are a few manufacturers who embrace the hobbyist community and offer their devices unlocked – the Nexus series of phones come with easily unlockable bootloaders.

Hardware

The hardware platform for Android is usually ARM (v7 and v8-A series), but Intel processors have also been used in Android devices since 2012. The Android system is built to run on both an x86 64-bit system as well as on the ARM64 platform.

One good thing about Android is that it doesn't have high RAM requirements. In fact, most Android devices need no more than 1 GB of RAM to run the OS smoothly. As of Android 7.1, most entry-level devices have 2 GB of RAM. In terms of display APIs, Android supports OpenGL and Vulkan, so a GPU is pretty much always present on an Android device.